Nagar YuwakShikshanSanstha's

YeshwantraoChavan College of Engineering

(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)

(Accredited 'A' Grade by NAAC)

Hingna Road, Wanadongri, Nagpur - 441 110

Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu

E-mail : principal@ycce.edu, info@ycce.edu

Form No YI-04

Department of Electronics Engineering

STUDENT'S DIARY/ LOG

Your Name: Khushi Narendra Atara

Your Institute: Yeshwantrao Chavan College of Engineering Nagpur

Company where internships are carried out : Cognizant Technology Solutions Pvt. Ltd

Period: From: 31/01/25 To: 13/05/26

Week: 9		DATE: From	30/03/26 to 03/04/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – Selenium WebDriver with POM, Apache POI & Mini Project (Days 33–39)	

• Studied Page Object Model (POM) design pattern: structure, benefits, separation of test logic from page elements.

• Learned Page Factory Model: @FindBy annotations, initElements() method, and creating reusable page classes.

• Practiced Apache POI: Excel Read Operation, Excel Write Operation, and Applying POI with POM for Flight Reservation.

• Understood Data-Driven Testing using Apache POI – reading test data from Excel sheets for parameterized test execution.

• Studied Selenium DatePicker using Actions Class and Calendar-based automation scenarios.

• Completed mandatory Tekstac hands-on: Excel Read Operation, Excel Write Operation, Applying POI – Flight Reservation, POM with Page Factory Model.

• Completed BU Chapter 12 hands-on (Apache POI) and Chapter content for Selenium WebDriver commands (Chapters 7–11).

• Completed Selenium WebDriver with POM and ApachePOI Code Challenge Assessment – 2 hrs, 3 attempts.

• Submitted Integrated Project Deliverable: Refactored Automation Framework (POM Implementation) and Data Management Strategy (Apache POI) – due Day 41.

• Continued Mini Project integration – automated individual functional scenarios and submitted for SME review.

Signature and stamp of Industry Supervisor

Nagar YuwakShikshanSanstha's
YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
(Accredited 'A' Grade by NAAC)
Hingna Road, Wanadongri, Nagpur - 441 110
Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu
E-mail : principal@ycce.edu, info@ycce.edu

Department of Electronics Engineering

Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

1. Company Overview:

Cognizant Technology Solutions is a leading global IT services and consulting company headquartered in the USA, with a major delivery presence across India. It specializes in digital transformation, cloud computing, AI, automation, software engineering, and quality assurance. Cognizant's QEA (Quality Engineering and Assurance) division focuses on intelligent and automated quality assurance, modernisation assurance, and experience assurance for enterprise clients across healthcare, finance, and technology sectors.

2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 9, I interacted with Cognizant Academy trainers and fellow GenC trainees from the QEA Selenium track. Trainer-led Master Sessions covered POM design pattern, Apache POI data-driven testing, and advanced WebDriver techniques for Days 33 to 39. All interactions were within the Training division, with additional engagement during the Code Challenge Assessment on Day 39.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Page Object Model (POM) Implementation: Attended Master Sessions on the POM design pattern and Page Factory Model. Practiced creating separate Page classes with @FindBy annotations and initElements() in Eclipse, separating test logic from UI element definitions for better maintainability.

Activity 2 – Apache POI Data-Driven Testing: Learned to use the Apache POI library to read from and write to Excel files within Selenium test scripts. Practiced Excel Read and Write operations and integrated POI with the POM framework for the Flight Reservation scenario on the Tekstac platform.

Activity 3 – Mini Project Integration & Code Challenge: Submitted the Mini Project deliverables including automated test scripts for individual functional scenarios. Also completed the Selenium WebDriver with POM and ApachePOI Code Challenge Assessment on Day 39 to benchmark skills in these areas.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Initially faced difficulty managing file paths for Excel files in the Apache POI read/write operations across different system environments. Resolved this by using relative paths and the ClassLoader.getSystemResource() method as suggested during the Master Session. This reinforced best practices for writing environment-independent automation code.

Nagar YuwakShikshanSanstha's
YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
(Accredited 'A' Grade by NAAC)
Hingna Road, Wanadongri, Nagpur - 441 110
Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu
E-mail : principal@ycce.edu, info@ycce.edu

Department of Electronics Engineering

5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with the Page Object Model (POM) design pattern and its implementation using Page Factory in Selenium. Also explored the Apache POI library for Excel-based data-driven testing, and studied BU Training Content for Chapters 8 to 12 covering WebDriver Actions, Multiple Windows, WebDriver Waits, Selenium Automation Techniques, and Apache POI with their respective hands-on, PPT, and video learning reference content.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

Providing a pre-built sample POM project structure as a reference template at the start of the POM module would help trainees get started faster. Currently, building the project structure from scratch during learning takes extra time. Having a reference project to compare against would accelerate understanding and reduce errors.

7. What did you find novel/exciting/valuable during this period?

Implementing the Page Object Model for the first time and seeing how it transforms messy, hardcoded test scripts into a clean, organized, and reusable framework was the most exciting experience of this week. The concept of separating test logic from page elements using Page classes and Page Factory annotations gave a real sense of how professional automation frameworks are built in industry.

8. What according to you, can be of use to your institute and students and in what ways?

Teaching the Page Object Model as part of academic software testing or Java projects would be highly beneficial. POM is an industry-standard design pattern used in virtually every enterprise Selenium framework. Introducing students to design patterns in the context of test automation would bridge the gap between academic Java knowledge and its real-world application in QA engineering.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 9 were completed. The Tekstac hands-on exercises for POM and Apache POI, BU Chapter content (Chapters 7–12), Udemy Section 4 (Apache POI – Data Driven Testing), the Code Challenge Assessment, Integrated Project deliverables, and Mini Project integration were all completed within the scheduled time.

Khushi N. Atara

Name and Signature of Student.

Nagar YuwakShikshanSanstha's
YeshwantraoChavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)
(Accredited 'A' Grade by NAAC)
Hingna Road, Wanadongri, Nagpur - 441 110
Ph.: 07104-242919, 242623, Fax: 07104-242376, Website: www.ycce.edu
E-mail : principal@ycce.edu, info@ycce.edu

Department of Electronics Engineering